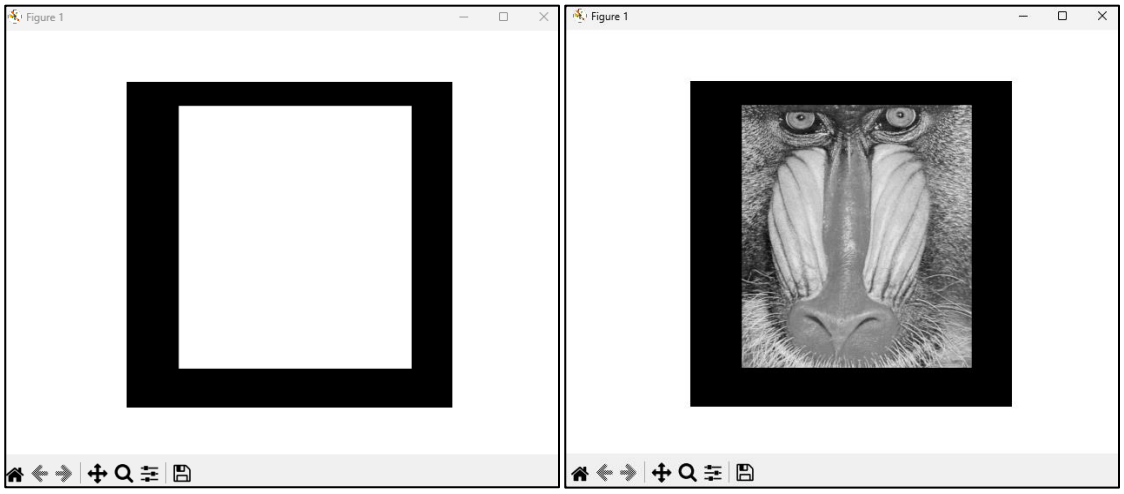
# Assignment 2

This is an individual assignment. This assignment is marked out of 5 points.

## Due Date: October 16 Friday, 2026, 11:59PM

1. Arithmetic operations between grayscale images:
   (a) [1 point] Write a function $imArithmetic4e(f1, f2, op)$ for performing operations $op$ on $f1$ and $f2$. Parameter $op$ is a character string that indicates the following arithmetic operations between $f1$ and $f2$: 'add' ($f1 + f2$), 'subtract' ($f1 - f2$), 'multiply' ($f1 * f2$), 'divide' ($f1/f2$). These are elementwise operations. Output image $g$ should be floating point. (*Hint*: Convert the input images to floating point to perform the arithmetic operations.)
   (b) [1 point] Read the image "BaboonGray.png" and use A1 function $mask$ and the 'multiply' option in $imArithmetic4e$ to highlight part of the baboon's face (shown below). This is an example of how to define a *region of interest* (ROI) in an image. (Hint: Use A1 exercise function of $cursorValues4e$ to get the coordinates needed to define the mask image.)




User-defined ROI

2. Brightness correction:
   (a) [1 point] Write a function $brightnessCorr(f, percent, op)$ to perform brightness correction on monochrome images. It should take as arguments a monochrome image $f$, a number $percent$ between 0 and 100 (amount of brightness correction, expressed in percentage terms), and a third parameter $op$ indicating whether the correction is intended to brighten ('brighten') or darken ('darken') the image.
   (b) [0.5 point] Read the image "cameraman.png" and brighten it by 30%. Show the result in a single plot.




3. Insert a small image into a large image:
   (a) [1 point] Write a function $insert(f1, f2, x, y)$ to insert a small image $f1$ into a large image $f2$ at the location of $(x, y)$. The function should return the new image.
   (b) [0.5 point] Insert "small_ubc_logo.jpg" into "cameraman.png" at location (10,10) and show the result.




## Requirements:

- Only the following packages are permitted: OpenCV-Python, Matplotlib, NumPy
- Only use the OpenCV-Python Arithmetic Operations
- Use Python version 3.10+

## Submission:

Submit everything as one zip file to Canvas.